

Learn HTML

Lesson 1: Introduction to HTML

Introduction to HTML – Study Notes

Key Concepts

1. **HTML (HyperText Markup Language)** – the standard markup language for creating web pages.
2. **Document Structure** – every HTML file follows a hierarchical skeleton: `<html>`, `<head>`, and `<body>`.
3. **Elements & Tags** – markup is written with opening and closing tags (e.g., `<p>...</p>`).
4. **The `<head>` Section** – holds meta information, title, links to CSS/JS, and other resources that are not displayed directly.
5. **The `<body>` Section** – contains all content that users see: text, images, links, forms, etc.

Summary

HTML is the backbone of the World Wide Web. It tells browsers how to structure and present content by using a series of tags that define elements such as headings, paragraphs, images, and links. Although browsers can render a page with just a few tags, following the proper document structure ensures consistency, accessibility, and easier maintenance.

A minimal HTML document starts with the `<!DOCTYPE html>` declaration, followed by the root `<html>` element. Inside `<html>`, the `<head>` element stores information about the page (its title, character encoding, external style sheets, scripts, etc.) that the browser uses but does not display directly. The `<body>` element follows and holds everything that appears on the screen—text, media, interactive controls, and more. Understanding this skeleton is the first step toward building well formed, functional web pages.

Important Points

- **Doctype:** `<!DOCTYPE html>` tells the browser to use HTML5 standards.
- **Root Element:** `<html>` encloses the entire document; it can include a `lang` attribute (e.g., `<html lang="en">`).
- **Head Contents:**
 - `<meta charset="UTF-8">` – sets character encoding.
 - `<title>` – text shown on the browser tab.
 - `<link rel="stylesheet" href="styles.css">` – attaches external CSS.
 - `<script src="app.js"></script>` – adds external JavaScript.
- **Body Contents:** Any visible HTML elements (headings, paragraphs, lists, images, etc.).
- **Well formedness:** Every opening tag should have a matching closing tag (except void elements like `<img>` or `<br>`).
- **Indentation & readability:** Use consistent indentation (2–4 spaces) to make the hierarchy clear.

Examples

Example 1 – Minimal HTML page

```
``html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My First Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is a simple HTML document.</p>
</body>
</html>
``
```

Explanation:

- The `<!DOCTYPE html>` declaration activates HTML5 mode.
- `<html lang="en">` sets the language to English.
- Inside `<head>`, the character set and page title are defined.
- `<body>` contains a heading (`<h1>`) and a paragraph (`<p>`), which are rendered on the page.

Example 2 – Adding an external stylesheet and image

```
``html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Styled Page</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <h2>Welcome to My Site</h2>
  
  <p>Enjoy exploring HTML basics.</p>
</body>
</html>
``
```

Explanation:

- `<link>` connects `styles.css`, allowing you to style the page separately.
- `<img>` is a **void element** (no closing tag) that displays an image; the `alt` attribute provides

alternative text for accessibility.

Quick Review

1. **What does the `<head>` element contain, and why isn't its content displayed on the page?**
2. **Name the three core tags that form the basic HTML document structure.**
3. **What is the purpose of the `<!DOCTYPE html>` declaration?**
4. **How do you specify the character encoding for an HTML document?**
5. **Give an example of a void element and explain how it differs from regular elements.**

Further Reading

- HTML5 Semantic Elements (`<header>`, `<nav>`, `<section>`, `<article>`, `<footer>`)
- Adding hyperlinks with `<a>` and creating lists (`<ul>`, `<ol>`, `<li>`)
- Introduction to CSS: styling your HTML content
- Accessibility basics: using `alt`, `aria`, and proper heading hierarchy
- Setting up a local development environment (VS /Code, Live Server extension)

